

Experiment name: Study of Switching Characteristics.

Objective: This experiment aims,

- to study the characteristics of the MOSFET.
- to study the MOSFET's behaviour as a switch.
- to study how gate voltage controls the conduction of MOSFET.
- to observe "Cut off" and "ON" states of MOSFET.

Theory:

MOSFETs (Metal Oxide Semiconductor Field Effect Transistors) are the most commonly used type of FET. They are kind of IGFET (Insulated Gate Field Effect Transistors), where the gate is separated from the channel by an insulating material. Usually, this insulator is silicon dioxide (SiO_2), which is why the name MOSFET is used. Like other IGFETs, a MOSFET has three main terminals. They are: Gate (G), Source (S), and Drain (D), sometimes a fourth terminal called the body or bulk is also present. In PMOS, the body terminal is connected

to the highest positive voltage in the circuit, while in NMOS it is connected to the most negative voltage.

There are four types of MOSFETs. They are:

1. Enhancement n-type MOSFET

2. Enhancement p-type MOSFET

3. Depletion n-type MOSFET

4. Depletion p-type MOSFET

The difference between them depends on whether the channel for current is already present or needs to be created when voltage is applied;

And whether the current is carried by electrons (n-type) or holes (p-type).

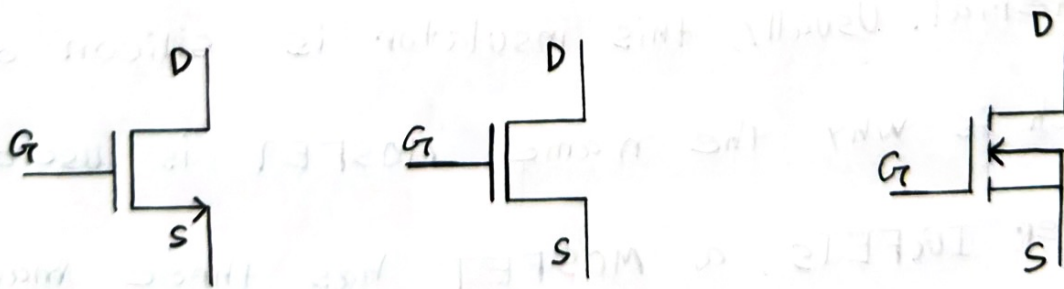


Fig: Symbols for Enhancement NMOS Transistor.

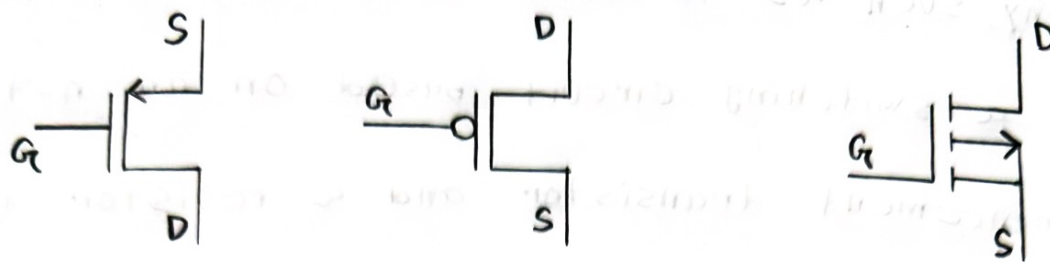


Fig: Symbols for Enhancement PMOS Transistor

When the drain and source are at ground and a positive voltage is applied to the gate, the positive charge pushes away the holes beneath the gate, creating an empty region there. At the same time, the gate attracts electrons into that area.

When there are more electrons than holes under the gate, an n-type channel forms, linking the source and drain. This channel is extremely thin, only a few nanometers wide, but all currents still flow through it.

MOSFETs are often used as switches in both analog and digital circuits. In analog systems, they can act like selectors, choosing one signal from

many, such as in data acquisition. A simple example of a switching circuit based on an n-type enhancement transistor and a resistor is shown below :

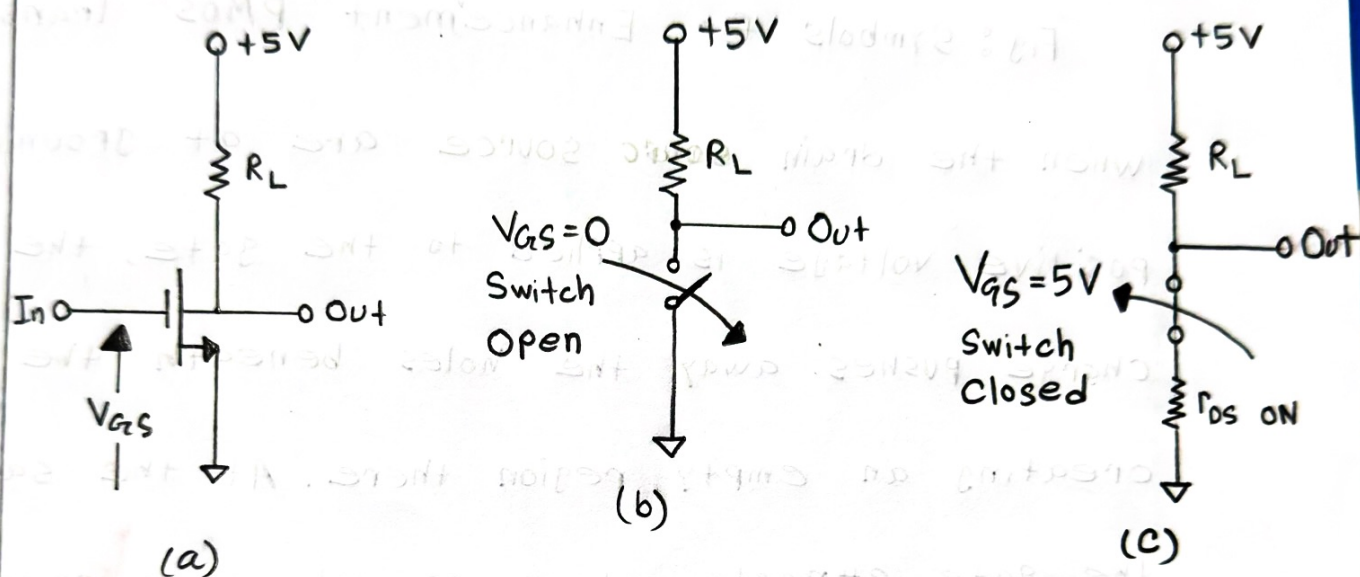


Fig: NMOS transistor switch

The gate voltage controls how easily current can flow through the MOSFET. When the gate voltage (V_{GS}) is zero or very small, the channel conducts poorly, so the transistor behaves like an open switch and no current flows through the load resistor.

When V_{GS} becomes higher than the threshold value, the channel conducts well and the transistor

works like a closed switch. Even though the resistance does not become zero, it becomes low enough that the output voltage becomes small. The figure above shows how an NMOS behaves like a switch for $V_{in} = 0$ (b) and for $V_{in} = +5V$ (c).

And, in PMOS transistor, it turns on only when the gate voltage is negative. If V_{gs} is zero, almost no current flows, so the output voltage becomes V_{DD} . But when V_{gs} is about $5V$, the transistor starts conducting, and the operating point shifts, causing V_{os} to drop to a low value. This behaviour makes the circuit work like an inverter, which is the basic building block of MOS logic gates.

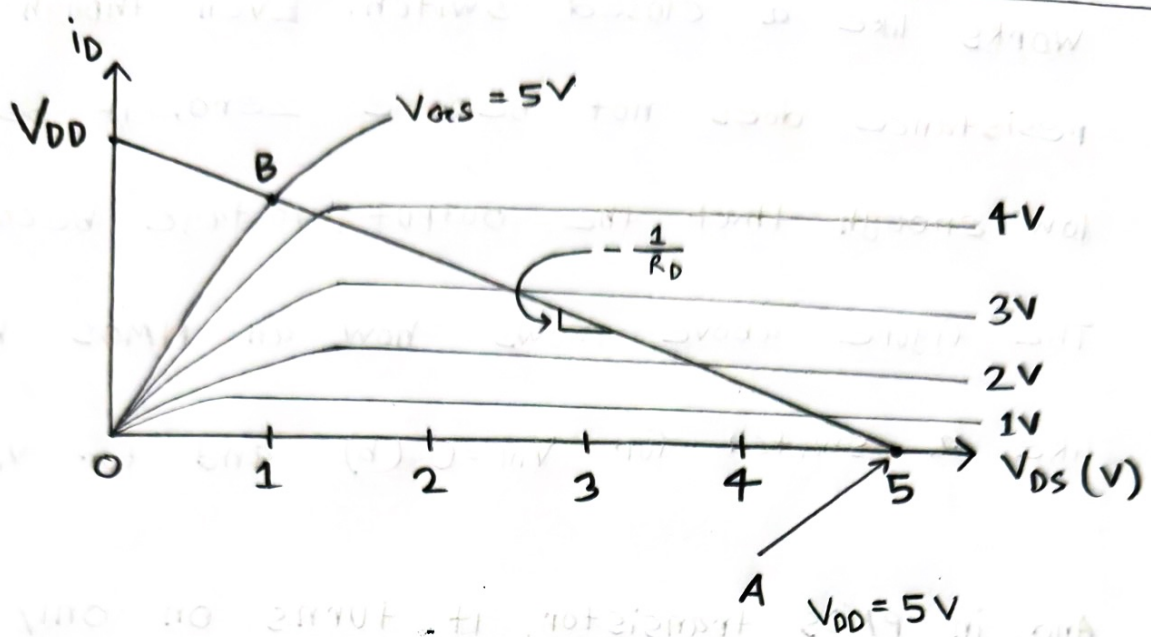


Fig : MOSFET switching characteristics

Equipment List:

Serial no.	Component Details	Specification	Quantity
1	MOSFET	IRF 540	1 Piece
2	Resistor	1 k Ω	1 piece
3	Trainer Board		1 unit
4	DC Power Supply		2 unit
5	Digital Multimeter		1 unit
6	Chords and Wire		As required

Circuit Diagram :

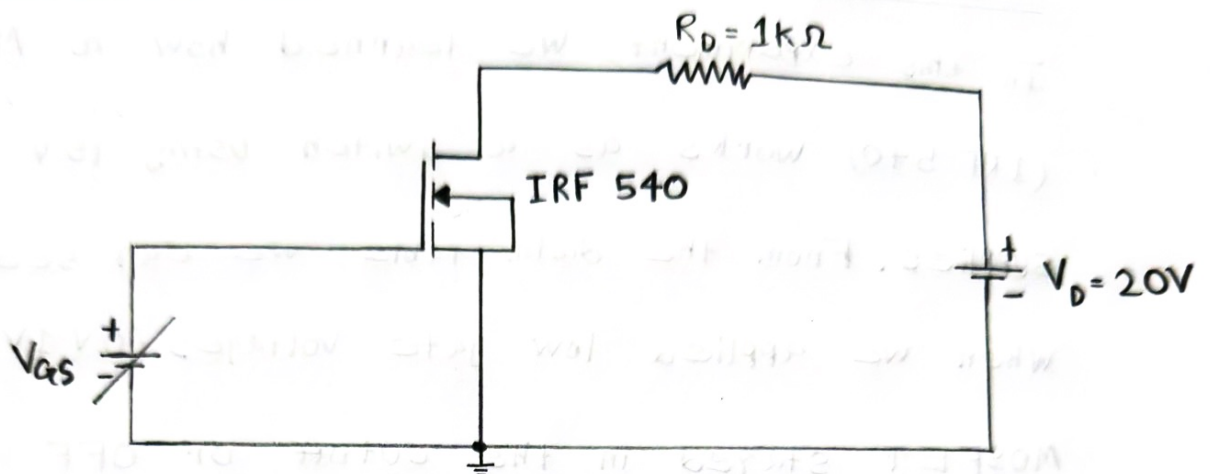


Fig : MOSFET Switching Circuit

Data Table :

$V_{DD} = 15V$				$V_{DD} = 20V$			
$V_{GS}(V)$	$V_{DS}(V)$	$V_L(V)$	$I_D(mA)$	$V_{GS}(V)$	$V_{DS}(V)$	$V_L(V)$	$I_D(mA)$
0	15.14	0	0	0	20.12	0	0
1	15.14	0	0	1	20.12	0	0
2	15.14	0	0	2	20.12	0	0
3	14.77	0.37	0.37	3	19.79	0.33	0.44
4	0	15.11	15.25	4	0	20.09	20.36
5	0	15.11	15.21	5	0	20.10	20.25
6	0	15.12	15.23	6	0	20.10	20.21

Discussion :

In this experiment, we learned how a MOSFET (IRF 540) works as a switch using 15V and 20V supplies. From the data table, we can see that, when we applied low gate voltages (0V, 1V, 2V), the MOSFET stayed in the cutoff or OFF state. And, no current flowed through the drain ($I_D = 0\text{mA}$); and the output voltage (V_{OS}) stayed almost equal to the supply voltage. This proves that at low gate voltage, the MOSFET behaves like an open switch and does not allow current to pass.

Then, around 3V gate voltage, the MOSFET began to turn ON, showing small drain current. As the gate voltage increased (4V, 5V, 6V), the device entered the fully ON state; where current increased and the output voltage (V_{OS}) dropped to 0V. This shows that higher gate voltage improves channel conduction and makes the MOSFET act like a

Closed switch.

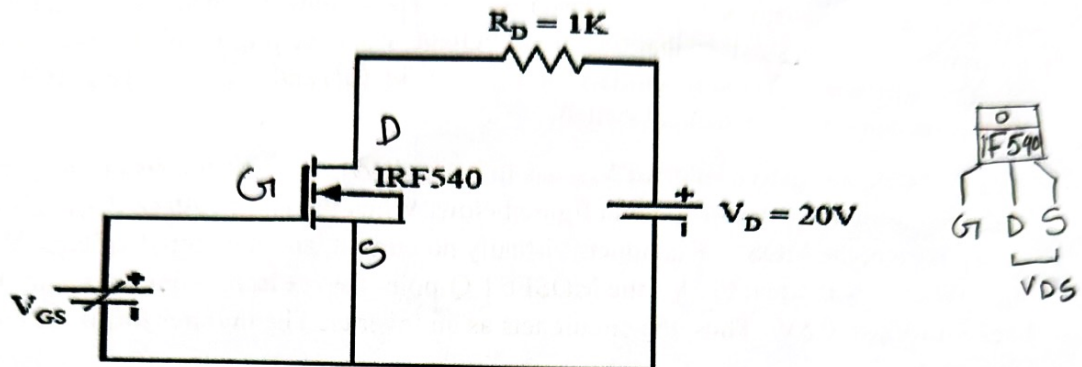
This experiment showed that a MOSFET can switch between ON and OFF states very quickly, proving its fast switching capability.

Although we faced some difficulties during the process, the results matched the theoretical concepts.

Overall, the experiment helped us understand how gate voltage controls the channel and how a MOSFET works as an electronic switch in real applications.

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Experimental Setup:



Procedure:

1. Set V_{GS} to zero and record the V_{DS} , V_L and I_D .
2. Increase the gate voltage V_{GS} gradually and record the readings.
3. Take reading until $I_D = 20\text{mA}$ (or the saturation current of the MOSFET).
4. Note the condition of V_{DS} and I_D of steps 1 and 3.
5. Repeat the experiment for $V_{DD} = 15\text{ Volts}$.

$R_{\text{practical}} = 0.989\text{ k}\Omega$

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2-12-2025

$V_{DD} = 15\text{V}$				$V_{DD} = 20\text{V}$			
V_{GS}	V_{DS}	V_L	I_D	V_{GS}	V_{DS}	V_L	I_D
0	15.14V	0V	0mA	0	20.12V	0V	0mA
1	15.14V	0V	0mA	1	20.12V	0V	0mA
2	15.14V	0V	0mA	2	20.12V	0V	0mA
3	14.77V	0.37V	0.37mA	3	19.79V	0.33V	0.44mA
4	0V	15.11V	15.25mA	4	0V	20.09V	20.36mA
5	0V	15.11V	15.21mA	5	0V	20.10V	20.25mA
6	0V	15.12V	15.23mA	6	0V	20.10V	20.21mA